Venkata Lakshmi Parimala Pasupuleti

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PROFESSIONAL SUMMARY

Machine Learning Engineer with 8+ years of production Python engineering across **full-stack development, ML systems, and affordable housing compliance automation** - all in the property leasing domain. Delivered LIHTC/Tax Credit, HUD, and Section 8/HCV compliance modules at Entrata across 100K+ unit portfolios. Built Leasel - an agentic RAG compliance system for LIHTC/Tax Credit (LangGraph agents, deterministic rules engine, precision/recall eval harness). Built AgentProbe - open-source multi-agent AI testing toolkit (LangGraph, ChromaDB RAG, 5-metric LLM-as-judge eval engine). M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

WORK EXPERIENCE

Entrata

Jan 2017 -- Aug 2024

Trainee → Associate Engineer → Software Engineer

- Sustained **99.5% uptime across 6+ months** for 100K+ monthly active users across 1M+ property records by engineering 20+ leasing and property management modules (Python, PostgreSQL, RabbitMQ) across 120+ Agile sprints.
- Shipped Entrata's **first production AI pipeline** across 1.6M real financial records - output validation, quality gates, and compliance controls at launch; later productized into a customer-facing feature (Python, PyTorch, OpenAI API).
- Automated **income eligibility verification, AMI threshold calculations, and recertification workflows** across 100K+ unit portfolios with zero compliance violations by building Section 8/HCV, LIHTC, and HUD modules.
- Cut microservices execution time **50%** by leading Docker containerization; improved query performance **30%** across 200+ SQL queries; achieved **100% PHI compliance** on AWS S3 migration with Libsodium-encrypted pipelines - zero security incidents.

George Mason University

Aug 2024 -- Aug 2025

Research Associate — Data Engineering · Statistical Modeling · ML Research

- Achieved **R^2 = 0.977** regression accuracy across 190+ countries by building multi-source prediction models (Python, R) on WHO and UK Road Safety datasets; surfaced anomalies manual review missed across 3 research pipelines by applying Isolation Forest.
- Improved research comprehension by **40%** across two cohorts by designing faculty leadership dashboards in Tableau, ggplot2, and Quarto.

PROJECTS

AgentProbe: Multi-Agent AI Testing & Evaluation Toolkit

Open Source

Python · LangChain · LangGraph · FastAPI · ChromaDB · OpenAI · Anthropic · Groq · Ollama · React · Docker · GitHub Actions — github.com/venkatalakshmiparimala/AgentProbe

- Delivered retrieval quality tracking, regression drift detection, and latency monitoring across concurrent agent test suites by building a **ChromaDB RAG pipeline** - directly applicable to leasing AI feature infrastructure.
- Blocked non-compliant agent runs before production by implementing a **5-metric LLM-as-judge eval engine** (factual accuracy, hallucination, retrieval, completeness, compliance) with CI/CD gates - multi-provider LLM routing (Anthropic, OpenAI, Groq, Ollama) across a 15-endpoint FastAPI backend.

Leasel: AI Lease & Affordable-Housing Compliance System

Python · LangGraph · SQLite · reportlab — github.com/venkatalakshmiparimala/Lease_complaince_rag

- Achieved **93.6% precision and 87.9% F1** across 10 compliance rules on 1,240 residential lease PDFs by building an agentic LangGraph RAG pipeline - pass@k harness (pass@1 81.7%, pass@3 88.3%) with regression gate; deterministic rules engine keeps compliance determinations fully auditable.

AI on FHIR: NLP-Driven Medical Records Query System

Flask · LLMs · FHIR · React · spaCy · BERT · Embeddings · Docker — github.com/venkatalakshmiparimala/AI_on_FHIR

- Reduced clinical retrieval time ~60% building an NLP platform translating natural language queries to FHIR APIs using LLMs, BERT, and semantic embeddings - **HIPAA-compliant RBAC** across structured medical records.

CrashClarity: Large-Scale Road Accident Analysis (104K Records)

Python · R · Scikit-learn · Random Forest · Logistic Regression · Feature Engineering — Live Dashboard

- Achieved **85%+ accuracy** on 104K+ UK accident records - Random Forest + Logistic Regression pipeline with feature engineering across weather, road, and time variables.

EDUCATION

M.S. Data Analytics Engineering — George Mason University — Graduated May 2026

GPA: 3.87 · Fairfax, Virginia

B.Tech Computer Science & Engineering — Jawaharlal Nehru Technological University, Anantapur — 2013 -- 2017

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, PL/SQL, R, Bash, HTML/CSS

ML & AI: PyTorch, TensorFlow, Scikit-learn, XGBoost, Random Forest, Isolation Forest, BERT, spaCy, LLMs, RAG Pipelines, LangChain, LangGraph, ChromaDB, Embeddings, NLP, Anomaly Detection, LLM Evaluation, LLM-as-Judge, Statistical Modeling, Model Deployment, MLOps, Prompt Engineering, OpenAI API, Anthropic API, FastAPI

Data & Visualization: NumPy, Pandas, Feature Engineering, Statistical Analysis, Jupyter, Data Visualization, Tableau, Power BI, ggplot2

Cloud & Infrastructure: AWS (S3, EC2, RDS, Glue), Azure, Docker, CI/CD (GitHub Actions), Microservices, Data Pipelines, ETL, RabbitMQ, PostgreSQL, MySQL, Oracle, Git, Linux, Agile/Scrum